

Multiclass Classification of Faulty Industrial Machinery Using Sound Samples

127176008 – Amirthavarthni G , 127176063-Manikandan S

Abstract

Reliable detection of faults in industrial machinery is essential for reducing equipment failures, production interruptions, and maintenance difficulties. Traditional monitoring approaches often depend on physical inspection, making fault diagnosis difficult when different machines and operating conditions are present. To address this challenge, this project proposes a machine learning based framework for multiclass classification of faulty industrial machinery using sound samples. The proposed system aims to identify both the type of industrial machine and its operating condition, namely normal or abnormal, from acoustic signals. The Malfunctioning Industrial Machine Investigation and Inspection (MIMII) dataset will be used as the dataset, containing sound samples from industrial machines under normal and abnormal conditions. Machine sound samples, such as fan and pump sounds, will be considered. The audio signals will first undergo preprocessing, followed by acoustic feature extraction using Mel-Frequency Cepstral Coefficients (MFCC) and spectral features. These features convert audio signals into numerical representations for machine learning. The extracted dataset will be divided into training and testing sets. Since normal and abnormal samples may be imbalanced, Synthetic Minority Over-sampling Technique (SMOTE) will be applied to the training data to generate minority samples and improve class balance. Z-score normalization will then transform the features to a common scale. Five supervised machine learning algorithms, namely Extremely Randomized Trees (ERT), Linear Discriminant Analysis (LDA), Gradient Boosting (GB), Light Gradient Boosting Machine (LGBM), and Support Vector Machine (SVM), will be investigated. The models will first classify machine sounds based on operating status and will then be evaluated for multiclass classification of machine type and status within a unified framework. Performance will be assessed using accuracy, precision, recall, F1-score, and confusion matrix. The proposed framework aims to provide a generalized sound-based fault detection approach for early identification of abnormal machine conditions and support predictive maintenance in industrial environments.

Keywords - Industrial machine sound, multiclass classification, machine learning, fault detection, predictive maintenance, MIMII dataset, MFCC, spectral features, SMOTE, Z-score normalization, SVM, LGBM.

References

Luana Gantert, Trevor Zeffiro, Matteo Sammarco, and Miguel Elias M. Campista, "Multiclass classification of faulty industrial machinery using sound samples," Engineering Applications of Artificial Intelligence, Volume 136, 2024, Article 108943.

GUIDED BY

Dr.Meena V